

A Comparative Simulation Study on State of Charge Estimation for 48V Lithium-Ion EV Battery Packs: Insights from CV and CC-CV Charging Techniques

Nivetha M

Department of Electronics and Communication Engineering
Amrita School of Engineering, Coimbatore
Amrita Vishwa Vidyapeetham, India
nivethamani1996@gmail.com

Dr. Ilango Karuppasamy

Department of Electrical and Electronics Engineering
Amrita School of Engineering, Coimbatore
Amrita Vishwa Vidyapeetham, India
k_ilango@cb.amrita.edu

Abstract—The battery management system is an essential and crucial part of Electric Vehicles. In the realm of Electric Vehicle (EV) applications, Constant Current Constant Voltage Charging (CC-CV) method has gained immense popularity for battery charging purposes. Li-ion batteries are the preferred option among rechargeable batteries due to their high energy density, reliable and dependability. Building upon the current analysis conducted during the constant voltage (CV) charging phase, this work presents a ground-breaking alternative of estimating the SOC of 48V battery pack using Coulomb Counting method during CC-CV charging to prevent overcharging and over discharging of the battery. One of the most important factors in determining the battery's capacity is SOC. The SOC estimated using the Coulomb Counting method during CV charging is compared with SOC estimated using the Coulomb Counting method during CC-CV charging. Battery pack of 48V is modeled using MATLAB Simulink. The average SOC obtained in CC-CV charging is 89% and in CV charging is 84%. The suggested approach extended battery life by enhancing the SOC and providing a healthy charging environment. The SOC estimation using the Coulomb Counting method during CV charging is implemented in the hardware using STM32 microcontroller.

Keywords—Battery Management System (BMS), Lithium-ion (Li-ion) battery, State of Charge (SOC), State of Health (SOH), Remaining Useful Life (RUL), Analog to Digital Converter (ADC), Coulomb Counting, Constant Voltage Charging (CV), Constant Current Constant Voltage Charging (CC-CV)

I. INTRODUCTION

A practical and cutting-edge approach to lower greenhouse gas emissions is through the use of EVs, which are quite simple to drive and do not require the fuel that conventional cars do. The following characteristics of EVs make them appear to be well suited for urban environments: they can drive frequently, they don't need gas stations, they don't store liquid, flammable fuels and don't emit emissions, and they produce the most torque at startup [1]. A key element of an electric vehicle is the rechargeable battery pack. These battery packs and systems are standardised under IS 17855: 2022, which is harmonised with ISO 12405-4: 2018. The test protocol for the fundamental characteristics of performance, dependability, and electrical functioning for battery packs and systems intended

for high-power or high-energy applications is included in this standard. This standard was developed with actual electric car scenarios in consideration. Multiple Li-ion cells are connected in series and parallel combination to form a battery pack. As lead-acid and nickel-based battery technologies become increasingly obsolete, lithium-based systems herald a new era in high-energy and high-power battery technology. Anode is graphite or silicon, cathode is lithium and non-aqueous electrolyte used in these Li-ion batteries. Its high energy density and high specific energy, combined with comparatively lighter weight, led to significant technological advancements. Due to Li-ion battery high heat potential and potential for fire and explosion, safety considerations when choosing a battery are essential. In order to control the battery safety requirements, several experiments have been conducted. [2].

Low temperatures have the effect of deteriorating Li-ion battery performance. Reduced power output, high polarization, and a decrease in the rate of diffusion are all observed at low temperatures [3].

II. LITERATURE SURVEY

BMS is a component of an intricate and quick-response power management system. The primary function of BMS is current and voltage monitoring, charge and discharge control, State estimation SOC, SOH, DOD, EOL and RUL, battery protection, control of power, thermal management, networking and communication, data gathering, data storage and diagnosis of fault [4].

The CC charging method charges the battery with a continuous current throughout the charging process. Challenges with CC charging include determining an appropriate charging current value that will meet the needs of capacity utilization and charging time. A high charging current speeds up charging but has a negative impact on the battery's aging process. Low charging current results in a very slow charge, which is inconvenient for EV applications though it offers high-capacity utilization.

CV charging method charges batteries by controlling a preset constant voltage. Its primary benefit is that it extends

battery life by avoiding over-voltages and irreversible side reactions. As the battery charges, the charging current drops because the voltage remains constant. Challenges with CV charging is determining the ideal voltage to balance electrolyte breakdown, charging speed, and capacity utilization.

The hybrid strategy CC-CV charging method blends the two charging techniques. Initially charges using the CC method during charging upon reaching the highest safe threshold value for voltage, the process transitions to the CV method. After the SOC level reaches the highest limit, the charging mode terminates. CC mode determines the charging time and capacity utilization determined by CV mode.

The SOC is the usable charge determined as a percentage of the maximum charge at a specific moment in time. This amount is typically computed by tracking the change in charge across the electrical load of the battery by integrating the current [5]. OCV method determines the SOC by summing up the battery's terminal voltage. A mathematical model or lookup table connecting OCV to SOC is needed, and this is usually acquired through calibration. Advantages are simple and inexpensive implementation. Disadvantage is accuracy affected by temperature factors. A small Alternating Current (AC) signal is applied, and the voltage response is measured by Electrochemical Impedance Spectroscopy (EIS) to determine the impedance (or resistance) of a battery at different frequencies, battery's state is determined by the impedance spectrum. By building a mathematical model from actual data and observations, as opposed to depending solely on basic physical principles, the empirical approach to battery modelling is achieved. A network of electrical components that approximates the dynamic processes taking place within the battery is used to create an Equivalent Circuit Model (ECM), a simplified description of a battery's electrical behaviour. To estimate the SOC, the recursive estimation algorithm Extended Kalman Filter (EKF) integrates measurements from several sources (such as voltage, current, and temperature) with a dynamic battery model. It is resilient against noise and measurement mistakes and offers real-time SOC estimate. Data-driven methodologies for estimating the SOC depend on statistical methods and machine learning algorithms to infer a battery's SOC from measured data, including temperature, voltage, current, and other pertinent characteristics. These techniques train models on past data so they can forecast a battery's SOC in real time or over time. Artificial Neural Networks (ANN) are able to acquire knowledge from training data about the intricate connections between battery characteristics (such as voltage, current, and temperature) and SOC. Convolutional neural networks (CNN) involves leveraging the spatial dependencies in input data to predict SOC values. Long short-term memory (LSTM) uses the temporal dependencies in sequential data to forecast future SOC values by using measurements from the past. Time-series prediction tasks are a good fit for LSTM networks, a form of Recurrent Neural Network (RNN) that is specifically made to handle sequence data with long-range dependencies. Based on characteristics taken from battery measurements, it can be trained to estimate

the SOC. Disadvantage is needs a lot of processing power and training data. Insufficient clarity and comprehensibility in relation to conventional techniques. [6].

SOH is a metric parameter that measures the reduction in performance brought on by aging batteries. It can be expressed as the battery capacity divided by the total maximum capacity of the current [7].

The BMS employs several techniques including impedance spectroscopy, Kalman filtering and Coulomb Counting to estimate a battery's the SOC. Model-based electro-chemical methods, data-driven methods, internal resistance measurement, and hybrid techniques that combine several techniques can be used to estimate SOH and RUL [8].

Cell balancing refers to the process of achieving equal voltage and the SOC between individual cells. Variation of voltage, temperature, and capacity balancing is required to provide greater capacity for the application [9] [10] [?].

The amount of time needed to recharge the battery is greatly decreased with fast charging. Because fast charging puts the battery cells under more chemical stress and increases heat generation, it can speed up battery degradation. Over time, this can cause the battery's capacity and lifespan to decrease [11]. A battery's lifespan increased by slow charging it normally puts less strain on the battery cells, which causes slower degradation. For drivers who must travel long distances or have limited time for charging, slow charging may not be the best option because it takes longer to fully recharge the battery [12].

The thermal management unit's primary function is to maximize output efficiency by either heating or cooling the battery. By directing cooling liquid towards the battery cell and if the battery becomes too hot, it may even cut the current. To warm up the battery pack for certain applications such as EVs this unit uses a heater when the temperature drops significantly [13].

In solid-state batteries, the liquid electrolyte found in Li-ion batteries is substituted with a solid electrolyte substance, usually made of polymer or ceramic. Benefits of solid-state electrolytes include increased stability, safety, and compatibility with high-energy electrode materials like lithium metal. The high interfacial electrode-electrolyte impedance, mechanical frailty, high operating temperature, and unstable cyclability are all drawbacks of solid-state batteries [14].

For Li-ion batteries, supply chain and material sourcing entails controlling the acquisition of components, raw materials, and manufacturing procedures to guarantee the effective and dependable production of batteries. They require a variety of components, including as graphite, lithium, cobalt, nickel, manganese, copper, aluminium, and electrolytes. The development of the EV business depends on maintaining a reliable and secure supply chain. Concerns exist about the origins of these metals as well; for instance, mining activities and the excessive water used in Li extraction have an adverse environmental impact [15].

The scope of work is restricted to comparison of the SOC estimation using CV charging and CC-CV charging.

III. BATTERY MODELLING

The model of 48V 80Ah Li-ion battery pack is simulated using MATLAB. Table I shows the specification of the battery pack. The battery pack charging current is 80Ah and discharging current 160Ah. Fig. 1 depicts the 12S4P Li-ion battery model. Battery is charged using two methods CV and CC-CV. Estimated SOC of the pack compared during the two charging patterns.

TABLE I
BATTERY PACK SPECIFICATION

Parameter	Specification
Battery Voltage	48V
Battery Capacity	80Ah
Charging Current	1C
Discharging Current	2C
Load Resistance	0.95 ohm



Fig. 1. MATLAB Simulink model of 48V Battery Pack

IV. CONSTANT VOLTAGE CHARGING METHOD

A. Coulomb Counting

The Coulomb Counting is an extensively used conventional technique for determining the SOC of a Li-ion battery. This method determines the battery's precise SOC based on the initial SOC and known specific battery capacity values. In this method, the remaining capacity and the cell's state of charge are calculated by adding the amount of the direct parameter of the discharging current [16] [17].

$$\text{SOC}(\text{final}) = \text{SOC}(\text{initial}) + \frac{1}{Q} \int_{t_0}^t I(t) dt \quad (1)$$

- SOC(final) is the state of charge at time t ,
- SOC(initial) is the initial state of charge,
- Q is the battery capacity,
- $I(t)$ is the battery current as a function of time,

- t_0 is the initial time, and
- t is the current time.

B. SOC Estimation using Coulomb Counting with CV Charging

The SOC estimated using the Coulomb Counting method is implemented in 48V battery pack. Fig. 2 depicts the Coulomb

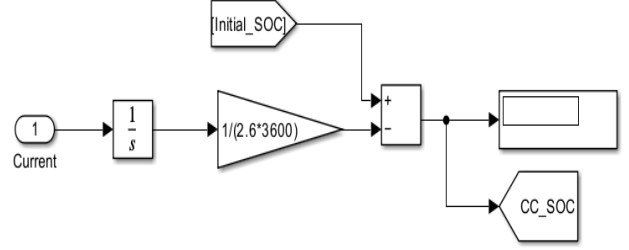


Fig. 2. The SOC estimation using Coulomb Counting during CV Charging

Counter in which input discharging current is integrated for a period of an hour. The SOC obtained by integrating the discharging current is deducted from initial SOC. Final SOC is obtained during the discharging process.

C. Simulation Results of the SOC estimation using Coulomb Counting during CV charging

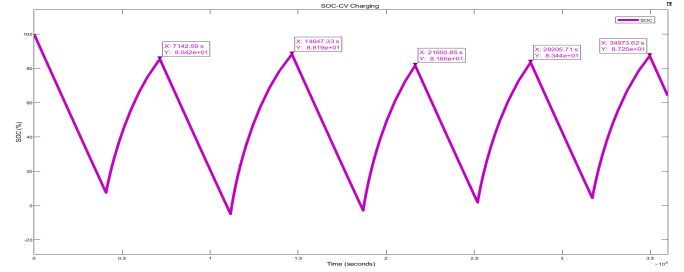


Fig. 3. Simulation results of the SOC during CV charging

Fig. 3 depicts the SOC curve of 48V battery pack during CV charging phase for period of about 36000s. The SOC rises due to CV charging and fall during discharging by using resistive load. The average SOC obtained during CV charging phase is 84.8%.

Initial charge of the battery was 100%. The SOC estimation depends on the measurement accuracy of initial charge of the battery.

Fig.4 depicts the current of 48V battery pack during CV charging phase for period of about 36000s.

Fig. 5 depicts the voltage of 48V battery pack during CV charging phase for period of about 36000s.

V. CONSTANT CURRENT CONSTANT VOLTAGE CHARGING METHOD

A. CC-CV

CC-CV process starts with CC battery charging with the nominal charging current. Once the battery reaches the max-

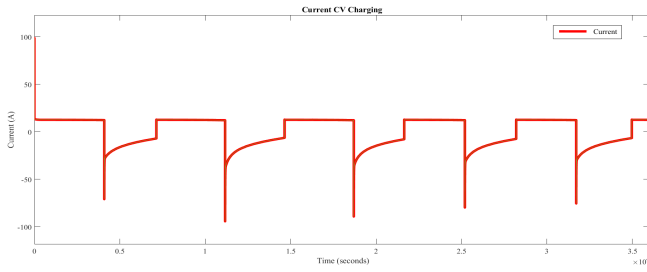


Fig. 4. Estimation of current during CV charging

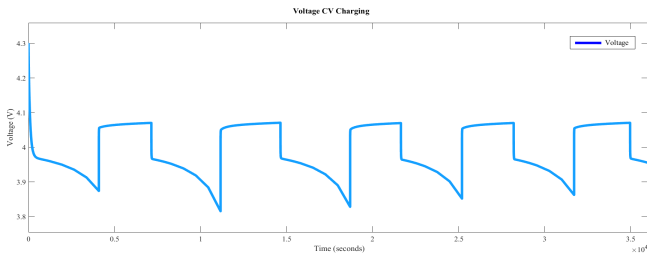


Fig. 5. Estimation of voltage during CV charging

imum threshold voltage of 48V the charging switches to CV charging. [18] [19] [20]. The experiment uses a constant current charging method, ranging from 20% to 90% SOC, to charge and discharge ternary li-ion batteries. The main reason for doing this is to take possible safety concerns into consideration when charging, allowing for a charge of up to 90% SOC.

B. The SOC Estimation using Coulomb Counting with CC-CV Charging

Initial battery charges by CC charging method with constant current of 1C rate when the battery cell voltage reaches the threshold limit of maximum voltage of 48.5V control logic is performed by the PI controller switches CC charging to CV charging. Fig. 6 depicts CC-CV charging module in which the charging is enabled when the SOC drops below 20%.

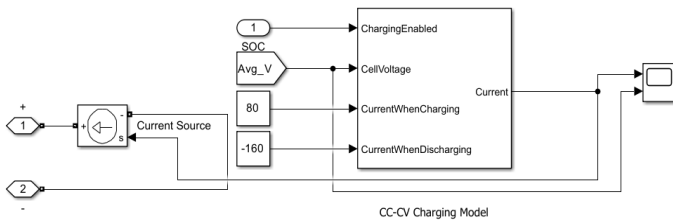


Fig. 6. Block Diagram of CC-CV Charging

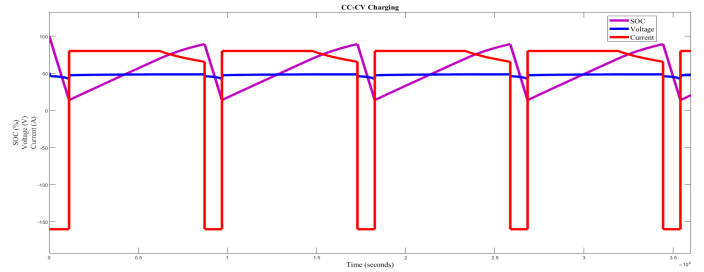


Fig. 7. Estimation of current, voltage and the SOC during CC-CV charging

C. Simulation result of the SOC by Coulomb Counting with CC-CV charging

Fig. 7 depicts charging current, voltage and SOC curve during CC-CV charging process for a period of about 36000s.

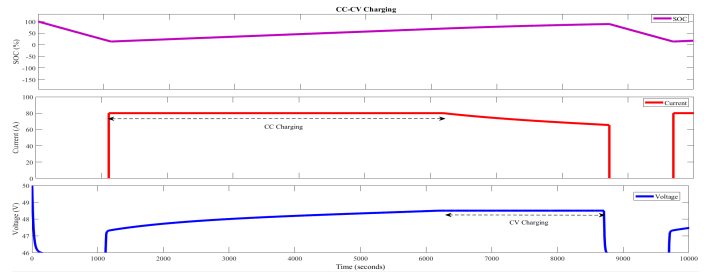


Fig. 8. Estimation of current, voltage and SOC during CC-CV charging

Fig. 8 depicts charging current, voltage and SOC curve in the course of CC-CV charging process for a period of about 10000s. Battery is charged by constant current and voltage rises during CC charging up to 6300s. Once voltage attained threshold of 48.5V at 6300s then constant voltage charges the battery pack upto 8700s. Battery starts discharging once SOC of 89% is attained.

Fig. 9 depicts the SOC curve of 48V battery pack during CC-CV charging phase for period of about 36000s. The SOC rises during charging and fall during discharging by using resistive load. An average SOC of 89% is obtained during CC-CV.

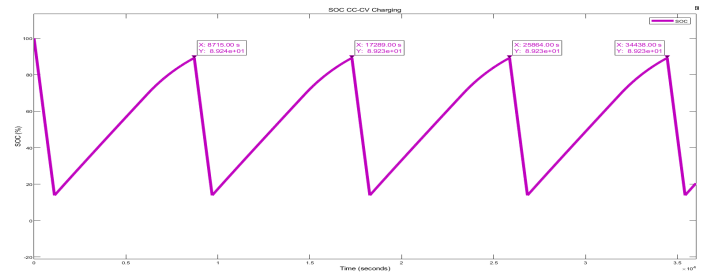


Fig. 9. The SOC estimation using Coulomb Counting during CC-CV Charging

Fig. 10 depicts battery current curve during CC-CV charging process for a period of about 3600s. During charging battery pack maximum current is 7A. During discharging battery pack maximum current is 52A.

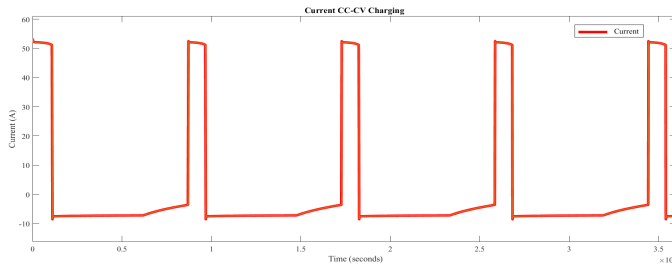


Fig. 10. Estimation of current during CC-CV charging

Fig. 11 depicts battery voltage curve during CC-CV charging process for a period of about 3600s. During charging battery pack maximum voltage is 4V and the minimum voltage is 3.5V.

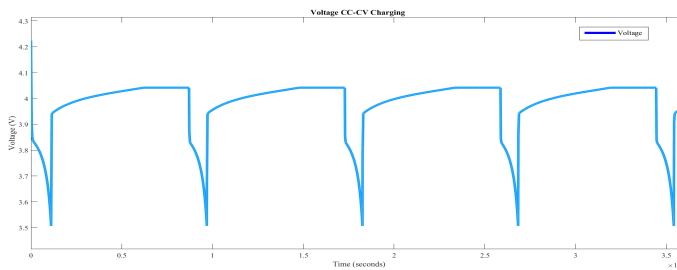


Fig. 11. Estimation of voltage during CC-CV charging

VI. RESULTS AND INFERENCES

An accurate 48V battery pack simulation model has been produced. Simulation models of the SOC estimation of Coulomb Counting with CC-CV charging and simulation model of SOC estimation of Coulomb Counting during CV charging are compared.

From the Fig. 3 and Fig. 9, it is observed that the Coulomb Counting with CC-CV charging has improved SOC of 89% is constant whereas in Coulomb Counting with CV charging has varying SOC of maximum 88% and minimum 81%. From the results it is inferred that Coulomb Counting with CC-CV charging shows an average improved SOC accuracy of 4.2%. CC-CV charging speeds up the charging process and during discharging process it improves the SOC of the battery.

PI controller is used to improve the performance of the battery. The error rate of the SOC is reduced by the proposed approach, and it improves the overall performance of the battery by estimating the accurate the SOC by coulomb counting.

VII. COULOMB COUNTING HARDWARE IMPLEMENTATION

Fig. 12 depicts Hardware setup of estimating the SOC using Coulomb Counting. Li-ion battery cell of 3.7V is connected to the sensors. Load of 10W resistive load is connected. The SOC variation in Li-ion cell is observed during discharging phase. The sensors monitor the battery parameters and performs the Coulomb Counting logic using MATLAB Simulink in the

STM32 microcontroller. Input parameters of battery measured by B25 Voltage sensor connected in parallel and ACS712 current sensor connected in series.

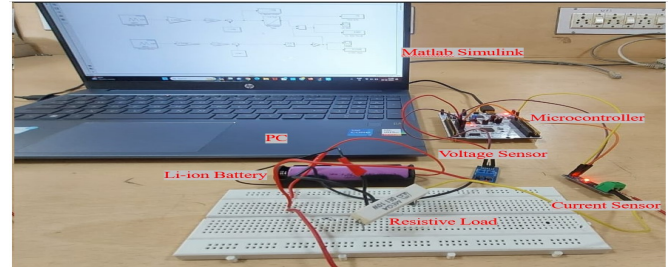


Fig. 12. Hardware Setup of Coulomb Counting

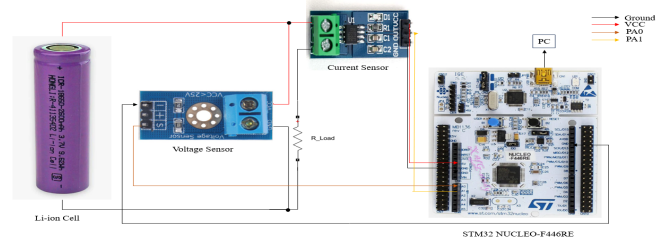


Fig. 13. Schematic Diagram of Coulomb Counting Hardware

Fig. 13 depicts the schematic diagram for the hardware method of Coulomb counting, wherein the STM32 microcontroller receives a measuring parameter signal from voltage and current sensors connected in a series and parallel configuration.

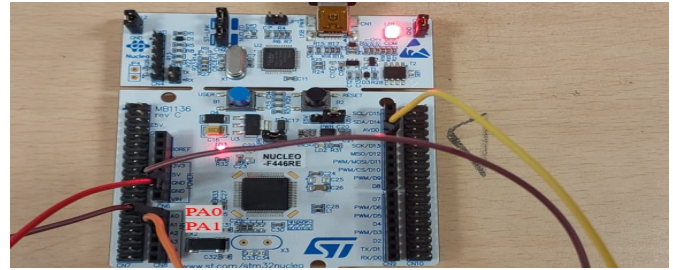


Fig. 14. Connection of STM32 NUCLEO- F446RE board

Fig. 14 depicts the output analog signal of sensor modules connected to the ADC pins PA0 and PA1 of STM32 NUCLEO-F446RE board. STM32 NUCLEO-F446RE board is connected to the Host PC and performs the Coulomb Counting logic by MATLAB Simulink. The obtained charge is then deducted from initial charge to give the final SOC of the battery.

The hardware implementation is carried out for the Coulomb Counting method during discharging process by connecting resistive load of 10W. Nominal voltage of Li-ion cell is 3.7V 2.6Ah. Fig.15 depicts the reduction of the SOC in the Hardware setup of the Coulomb Counting during discharging phase when the battery with initial voltage of

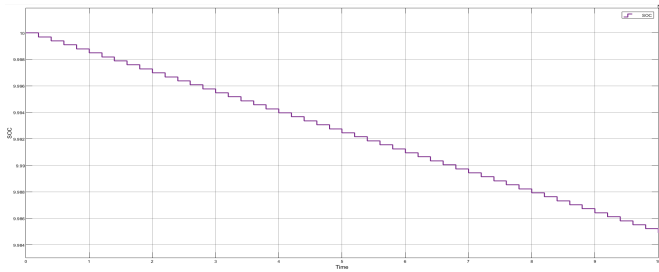


Fig. 15. The SOC estimated by Hardware Setup of Coulomb Counting

3.1V is connected to a load of 10W. The SOC reduces during discharging phase from 10% to 9.985% for about simulation period of 10s. The drop in SOC of about 0.015% is observed in hardware setup.

VIII. DISCUSSIONS AND CONCLUSION

The objective of creating an effective 48V battery pack with improved performance and life cycle for electric vehicles has been carried out. The accuracy of the SOC estimation can be increased by improving the battery testing system's performance by lowering measurement noise. Data-based SOC estimate in EVs can be implemented more effectively with the use of cloud computing and IoT technologies in real-time applications. Average SOC estimated for charging and discharging of battery pack using Coulomb Counting during CV charging process is 84.8% and during CC-CV charging process is 89%. It is observed that the proposed method of estimating the SOC using Coulomb Counting during CC-CV charging has an improved accuracy of 4.2%. By taking into account for the specific characteristics of Li-ion batteries, Coulomb Counting in conjunction with CC-CV charging provides increased accuracy throughout the charging process. 48V 80Ah Li-ion battery pack is used in applications such as E-Vehicles, RV, Golf, Solar, Off Grid, Propulsion. In order to improve the efficiency and the lifetime of the battery pack it is necessary to charge the battery using Constant Current Constant Voltage Charging method. Future work can be done by improving the SOC accuracy by Coulomb Counting approach with CC-CV charging using numerical iteration approach.

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